

$$\Rightarrow v > \sqrt{gR}$$

11 D $E_i = E_f$

$$\frac{1}{2}mv_e^2 + \left(-\frac{GMm}{R}\right) = 0 + 0$$

$$v_e = \sqrt{\frac{2GM}{R}} \propto \sqrt{\frac{M}{R}}$$

12 C $W = \Delta U$